



Fig. 2: Glucometer that uses enzyme glucose oxidase to break blood glucose down

the transducer) baseline. The signal processing normally involves subtracting a reference baseline signal, derived from a similar transducer from the sample signal, amplifying the resultant signal difference and electronically filtering (smoothing) out the unwanted signal noise.

The relatively-slow nature of the biosensor response considerably eases the problem of electrical noise filtration. The analogue signal produced at this stage may be output directly but is usually converted to a digital signal and passed to a micro-



Fig. 3: Pregnancy test that detects hCG protein in urine

processor stage where data is processed, converted to concentration units and output to a display device or data store.

Advantages

Some advantages are:

1. Rapid, continuous measurement
2. High specificity
3. Very less usage of reagents required for calibration
4. Fast response time
5. Ability to measure non-polar

molecules that cannot be estimated by other conventional devices

Applications

1. Monitoring glucose levels in diabetic patients
2. Food analysis
3. Environmental applications
4. Protein engineering and drug-discovery applications
5. Waste water treatment

Future prospects

Trends in biosensor technology over the past 30 years have taken this equipment from a simple and cheap component to the integration of several sensor systems into one unit including multiple components, making these systems smaller and tailored for mass production. The vision for the biosensor industry is to create micro-scale technology that will be suitable for performing sample preparation, analysis and diagnosis all with one chip. ●

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